

Sexual Presence in Virtual Reality: a qEEG Exploration.

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Abstract

The increasing use of immersive technologies for sexual purposes raises questions about their capacity to enhance a unique aspect of presence—Sexual Presence (SP). Investigating this phenomenon hinges on our ability to measure it accurately. This paper improves our understanding of SP by identifying potential quantitative electroencephalography variables associated with SP. Twelve heterosexual cisgender males were exposed to virtual scenarios featuring sexual content performed by a Virtual Character (VC). After viewing, participants completed a Sexual Presence questionnaire. Correlations were observed between self-reported SP and the alpha band activity in the frontal and parietal regions.

Keywords: Sexual Presence, Virtual Reality, Quantitative Electroencephalography.

Index Terms: H.1.2 [Models and Principles]: User/Machine Systems—Human factors, Human information processing; H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities.

1 INTRODUCTION

Annually, the consumption of pornography, notably on platforms like Pornhub, is on the rise, with 42 billion visits recorded in 2019 [1]. The evolution of mediums for explicit content consumption is evident—from photography and magazines in the 1960s to the widespread accessibility of pornographic videos with the advent of the Internet in the 1990s. A new turning point is emerging with the surge in popularity of immersive technologies, particularly virtual reality (VR), for engaging with adult content. VR pornography is projected to become a billion-dollar industry by 2027 [2], especially among Generation Z viewers [1].

Given the established associations between traditional 2D pornography and problem related to sexual development and sexual behaviors [3], it becomes crucial to explore the potential effects of 3D immersive pornography on the user's mental well-being. In recent years, there has been a growing interest among researchers, particularly in understanding how VR pornography can influence users' subjective perception [4–6]. Research has indicated that, much like the overall sense presence [7], the enhanced immersion provided by VR pornography can lead to an amplified perception of a particular form of presence within this context, known as Sexual Presence (SP) [5]. SP is a complex phenomenon that, in the technologically mediated context of the actual paper, encompasses "the feeling of connecting with someone on a sexual level".

In this study, our goal is to delve into qEEG data to pinpoint markers that signal an enhanced sense of SP during engagement with immersive sexual content. Subsequent research endeavors

could utilize these markers to explore whether specific user characteristics predispose individuals to the development of an intensified SP feeling and, potentially, problematic sexual behavior. Additionally, these markers could provide guidance for VR developers in various areas where VR technologies are utilized in the field of sexuality, including applications in sexual aversion treatment [8] or the assessment of deviant sexual preferences [9]. In selecting markers for exploration, we gleaned insights from existing research on various forms of Presence that could be connected with SP. So far, the assessment of SP, primarily linked to sexual arousal, has been conducted using alpha asymmetry in the frontal region [10], already associated with reported sexual arousal [11]. Acknowledging that SP extends beyond sexual arousal, we recognize the need to explore potential markers associated with other forms of presence. Social Presence, corresponding to the "feeling of connecting with someone" [12] has EEG correlates investigated. Event-related research protocols revealed that social presence may be gauged through frontal activity at AFz and FPz electrodes [13]. Soiné et al. (2021) propose that these electrodes, sensitive to anterior cingulate cortex activation changes, are associated with emotional responses tied to the interpersonal connection inherent in social presence. This renders them a suitable choice for evaluating social presence and potentially, SP. Lastly, Spatial Presence, defined as the "feeling of being there", in the virtual environment, is associated with distinct brain activity patterns in the parietal lobe and dorsolateral prefrontal cortex, involving alpha and theta rhythms [14]. These could potentially serve as interesting markers for measuring SP.

2 EXPERIMENTAL OVERVIEW

This paper utilizes data collected from an extensive study involving diverse immersive conditions, wherein participants observed a VC engaging in sexually explicit gestures. The complete experiment comprised an initial exposure to baseline conditions featuring neutral or asexual scenes, succeeded by the presentation of four sexual conditions. Participants spent approximately 45 minutes in the virtual environment. After each condition, they utilized a controller to respond within the Head-Mounted Display (HMD), to a SP [10] questionnaire, individual answers being on a sliding scale from 1 to 100. Only data from the first sexual condition were utilized for the current study.

The virtual environment (VE) was modeled and rendered using the Unity game engine (Version 2020.3.36, Unity Software Inc.). The virtual character (VC) was sourced from the Genesis 8 collection (Daz3D, Daz Production Inc.). To animate the character, a motion capture session was carried out with a female performer using eight Prime13 cameras and Motive: Body (Version 2.3.1, NaturalPoint Inc). The animation was produced using MotionBuilder (Version 2019, Autodesk Inc.). The computer used to render the VR experience was equipped with an 8GB Nvidia GeForce RTX 3080 graphics card (MSI, Taiwan), and an Intel Core i7-10700K processor, along with 32 GB of RAM. We used the HTC Vive Pro Eye Head-Mounted Display (HMD) (HTC, Taiwan) as visualisation device.

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3 DATA RECORDING, PROCESSING AND ANALYSIS

EEG measurements were acquired using a 32-active electrode EEG cap following the 10-20 system (Acticap, Brain Vision). The EEG signal was amplified in real-time using the ActiChamp amplifier from Brain Vision and recorded with Brain Vision's MOVE and Recorder software (Version 1.20.0401, Brain Vision). The online reference was set at Cz. EEG data were sampled at a rate of 500 Hz and underwent online filtering, including a low-pass filter at 1.59 Hz, a high-pass filter at 70 Hz, and a notch filter at 60 Hz. The EEG data underwent preprocessing using Analyzer software (Version 2.1, Brain Vision). An ocular Independent Component Analysis (ICA) was applied for further artifact removal, using the meaned slope algorithm, the given condition's dataset, and the Infomax and sum of squared correlations methods as implemented in Analyzer. Subsequently, all channels were re-referenced to the mean of the mastoids; any channels that were considered noisy upon visual inspection were excluded from the analysis. Additionally, spontaneous, and isolated artifacts, not related to eye movement, were manually removed. The alpha (8Hz – 13Hz) band frequency was extracted using the FFT Band Export solution (BrainVision) with an epoch duration of two seconds and an overlap of 50%. Following this, we calculated the average power for each condition by considering all epochs in which the character was visible to the participant. Next, we normalized the data distribution by applying a natural logarithm. Subsequently, we computed a relative asymmetry score by subtracting the alpha power of the left hemisphere from that of the right hemisphere. Correlations were utilized to explore the potential relationships between self-reported SP and qEEG variables.

4 RESULTS AND DISCUSSION

For this preliminary exploratory analysis, we aimed for simplicity by including only 12 heterosexual, male, and cisgender participants (Age: 25.9 years (± 7.7)). This approach seeks to use a minimal number of variables to explain a maximum amount of variance. However, we acknowledge the importance of incorporating female participants into the research protocol, as well as recognizing the significance of sexual orientation and gender identity. We understand that this approach represents a simplification. Due to the bias that can induce multiple correlations, we restrict our interpretation to correlation equal or below 0.01.

	FP1	FP2	F7	F3	Fz	F4	F8
r	-0.72	-0.58	-0.60	-0.70	-0.75	-0.72	-0.68
p	0.009	0.052	0.051	0.013	0.005	0.008	0.014

Tab 1. Correlations between SP and frontal electrodes (alpha band)

	P7	P3	Pz	P4	P8
r	-0.74	-0.63	-0.65	-0.57	-0.59
p	0.007	0.027	0.022	0.055	0.043

Tab 2. Correlations between SP and parietal electrodes (alpha band)

	FP2-FP1	F8-F7	F4-F3	P8-P7	P4-P3	FP2 FP1 Fz
r	-0.19	-0.17	-0.13	0.59	0.41	-0.70
p	0.546	0.612	0.697	0.044	0.189	0.010

Tab 3. Correlations between SP and computed variables.

In the frontal region, as illustrated in Tab. 1, negative correlations emerge between total power in the alpha band and SP in FP1 ($r = -0.717$, $p = 0.009$), Fz ($r = -0.745$, $p = 0.005$), and F4 ($r = -0.723$, $p = 0.008$). Those negative correlations suggest that heightened SP could be associated with a reduction in alpha power, generally taken as indicating increased brain activity [11]. We examined relative asymmetry (Tab. 3) and, against expectations as anticipated for SP, found no correlation, [10, 11]. However, frontal region activity is also associated to social presence, especially in in AFz and FPz [13].

By averaging FP1, FP2, and Fz to compensate for the absence of AFz and FPz in our EEG setup, [13], we noted a correlation ($r = -0.700$, $p = 0.01$), suggesting that there might be a relation between Social and Sexual Presence. Regarding the parietal region, as shown in Tab. 2, a correlation with SP was observed in P7 ($r = -0.735$, $p = 0.007$). Furthermore, a marginal correlation was identified between SP and relative asymmetry (Tab. 3) in P8-P7 ($r = 0.588$, $p = 0.044$). Although modest, the positive correlation might suggest that a larger relative asymmetry corresponds to a higher level of SP. Considering that power suppression in alpha activity or desynchronization indicates heightened spatial presence [10], we may infer that the sensation of "being there" contributes to the development of SP. However, it is important to note that despite some strong correlations, the small sample size and multiple correlation necessitates caution in interpretation, and validation on a larger sample is required. Moreover, the lack in realism of the virtual character and of the animation could have led to a lower SP and less noticeable effect on markers.

5 CONCLUSION

Our research aimed to identify EEG correlates indicating a heightened sense of SP as individuals engage with immersive sexual content. In summary, our study unveils patterns of decreased alpha power within the parietal and frontal regions, correlating with an augmented self-reported SP and suggesting a potential connection with other well-known types of presence. However, future research endeavors could greatly benefit from pinpointing markers specific to SP, allowing for a clear differentiation from other forms of presence.

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